

**CSE4067**

# **Introduction to Blockchain Programming**

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# How to Become a Crypto-Billionaire

How a 26-Year-Old College Dropout  
Makes \$15,000 a Month With Bitcoin  
and Cryptocurrency Without  
Breaking a Sweat

Click here to open a new tab Published on February 16, 2021, [FinanceIndex.co](https://financeindex.co)



## Meet the bitcoin investors who got insanely rich off crypto

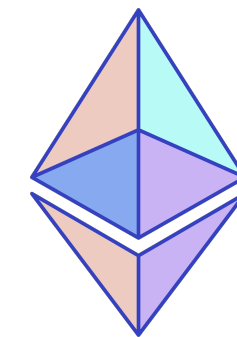
By [Suzy Weiss](#)

January 13, 2021 | 7:23pm | Updated



# What you will learn

- Fundamentals of Blockchain
  - (e.g. **Bitcoin, Ethereum**)
- Fundamentals of Cryptography
- Privacy in Blockchain
- Basics of Zero-Knowledge Proof
- Blockchain Programming



# What you will **not** learn

- How to become a crypto-billionaire (**for sure**)
- Crypto Trading
  - (e.g. day trading, mining, investment strategies)
- Illegal Crypto Activities
  - (e.g. cybercrime, money laundering, hacks)

# Languages

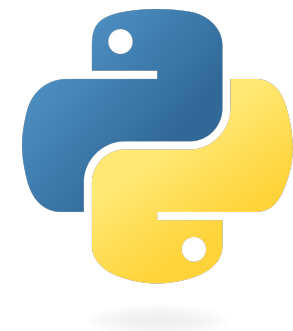
- **Solidity**



- **ZoKrates**



- **Python**



# Money, Money, Money!



# Properties of Money

- **A medium of exchange:** Can be used as medium for the exchange of goods
- **A unit of account:** Can be used for pricing of all goods and services, for accounting purposes and debt recording
- **A store of value:** Storing and retrieving it at a point in the future maintains its value.

# Money v1.0



# Money v1.0

- A medium of exchange: **Medium**
  - Face to face transactions
- A unit of account: **Mediocre**
  - Fungible, but not divisible well
  - Typically forgeable
- A store of value: **Bad**
  - Some objects may deteriorate.
  - May have unknown hidden quantities.

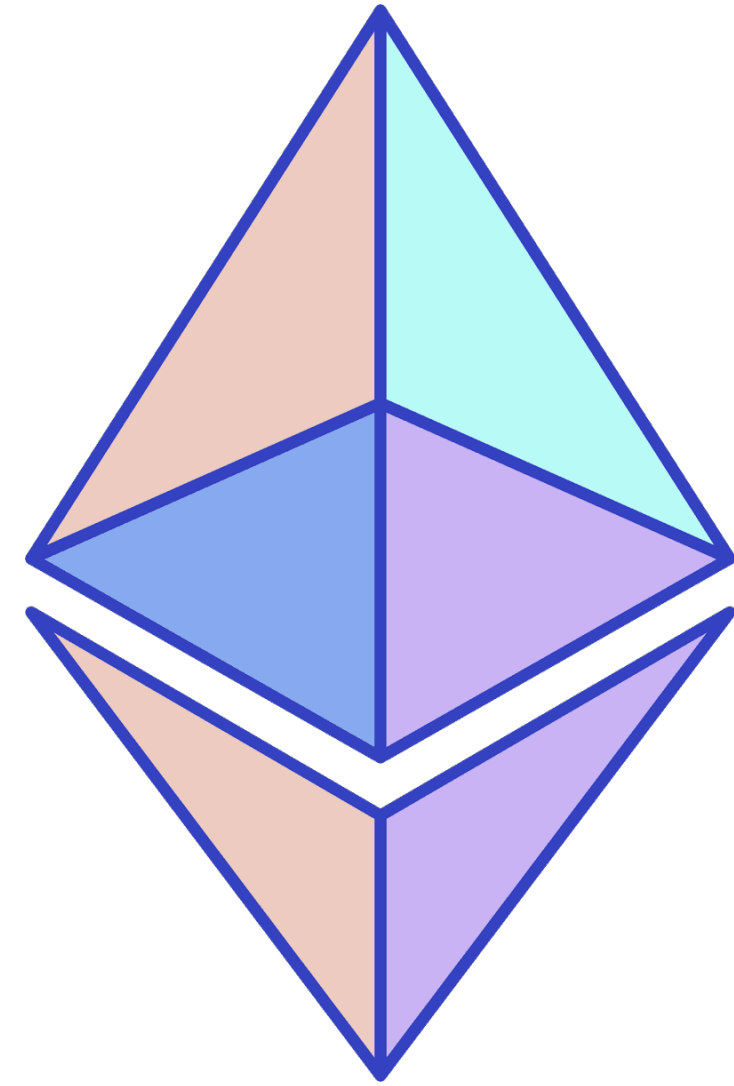
# Money v2.0



# Money v2.0

- A medium of exchange: **Good**
  - For transactions within the domain of the trusted entity
- A unit of account: **Great**
  - Fungible & divisible
- A store of value: **Mediocre**
  - Tied to the availability & reputation of the issuing entity

# Money v3.0



# Money v3.0

- A medium of exchange: **Improving**
  - Assuming internet connectivity / adoption
- A unit of account: **Good**
  - Fungible & divisible
- A store of value: **Good**
  - No trusted parties
  - No natural deterioration

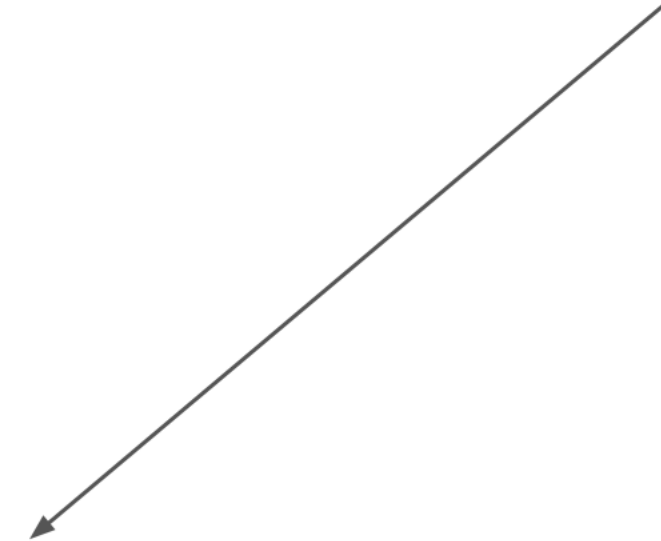
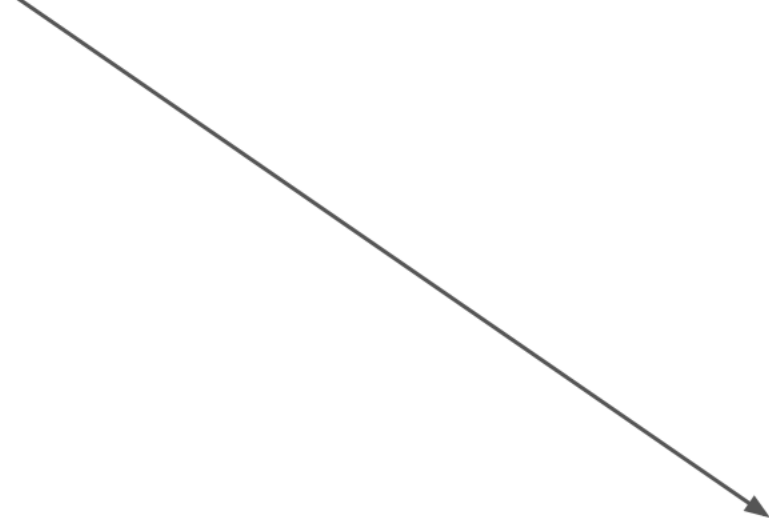
# Short Story: Never Ending Book

- Anyone can be an scribe (author) to produce a page.
- New pages are produced indefinitely as long as scribes are interested in doing so.
- Each new page requires some effort to produce.



# Never Ending Book: Consensus

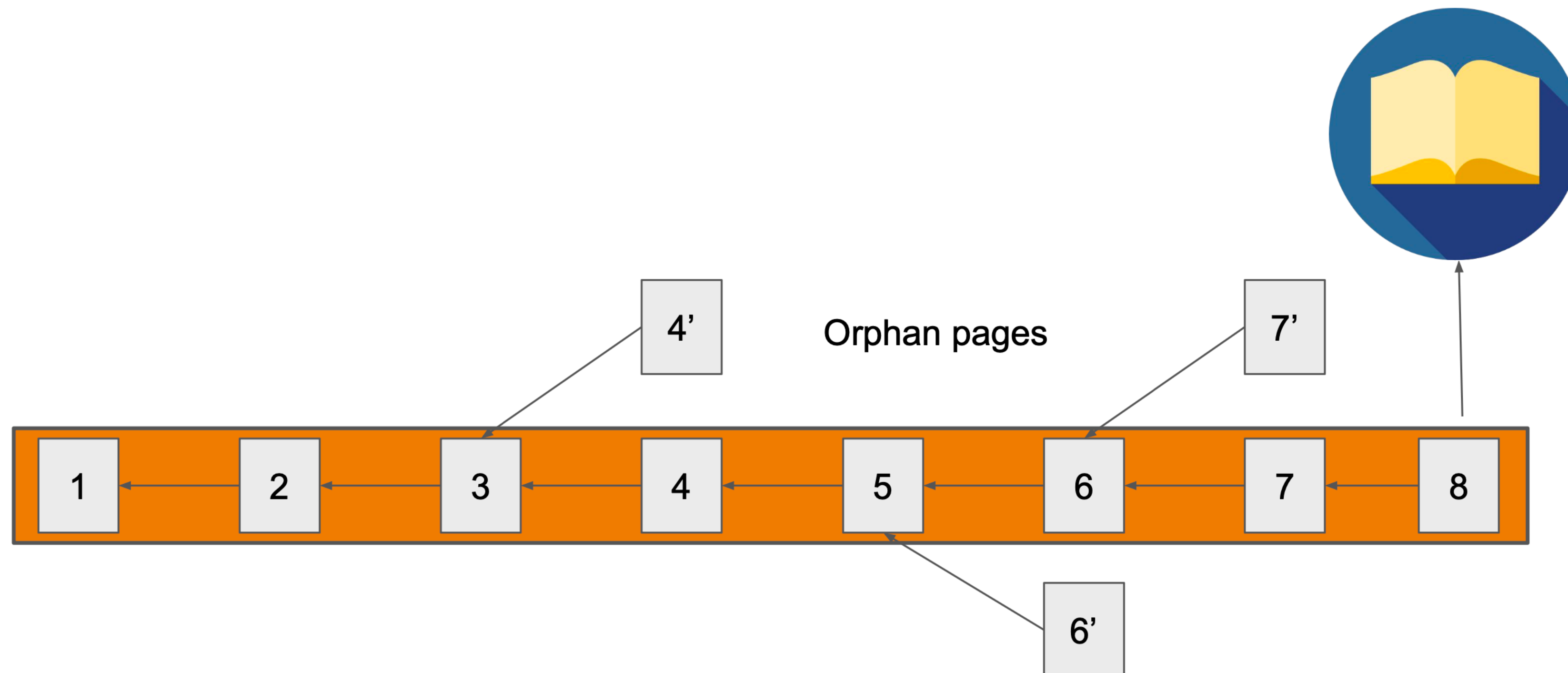
- If multiple conflicting books exist, which is the “right one” ?



- The **correct book** to work on is the book with the most pages. If multiple exist, just pick one at random.

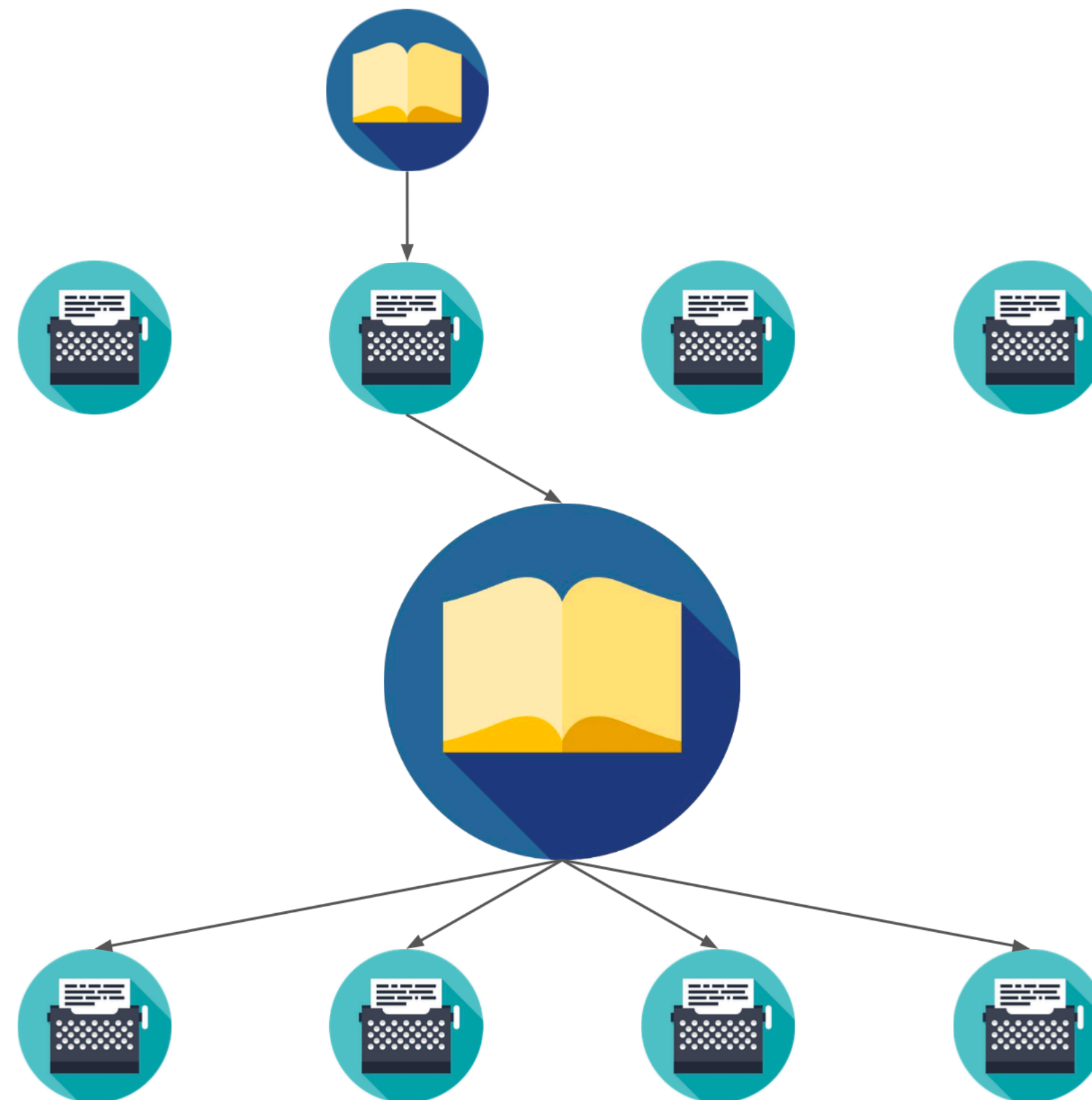
# Never Ending Book: Consensus

- Each page refers only to the previous one
- Current assembled by stringing together the longest sequence of pages



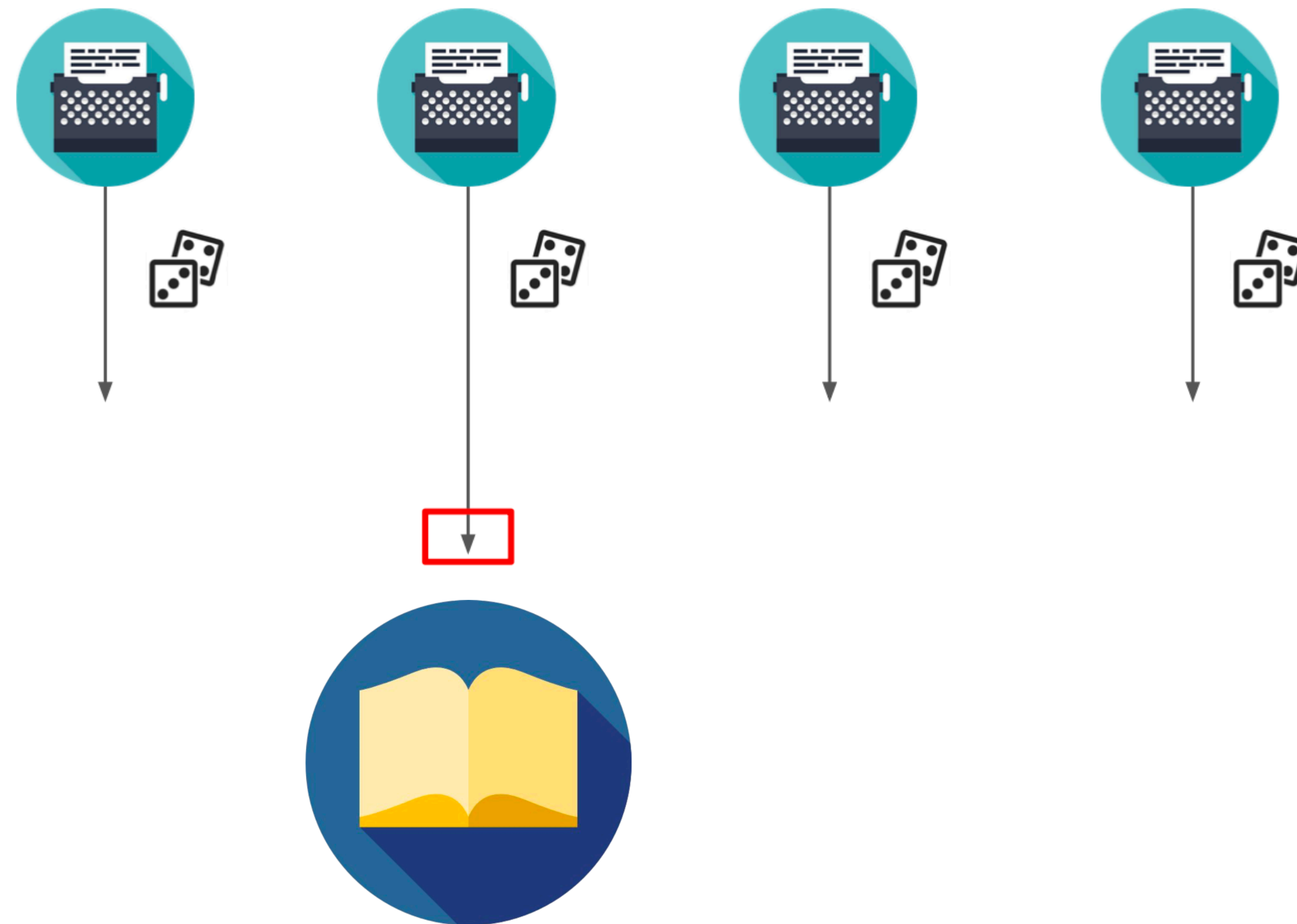
# Never Ending Book: Extending it

- The first scribe that discovers a page announces it to everyone else



# Never Ending Book: Effort

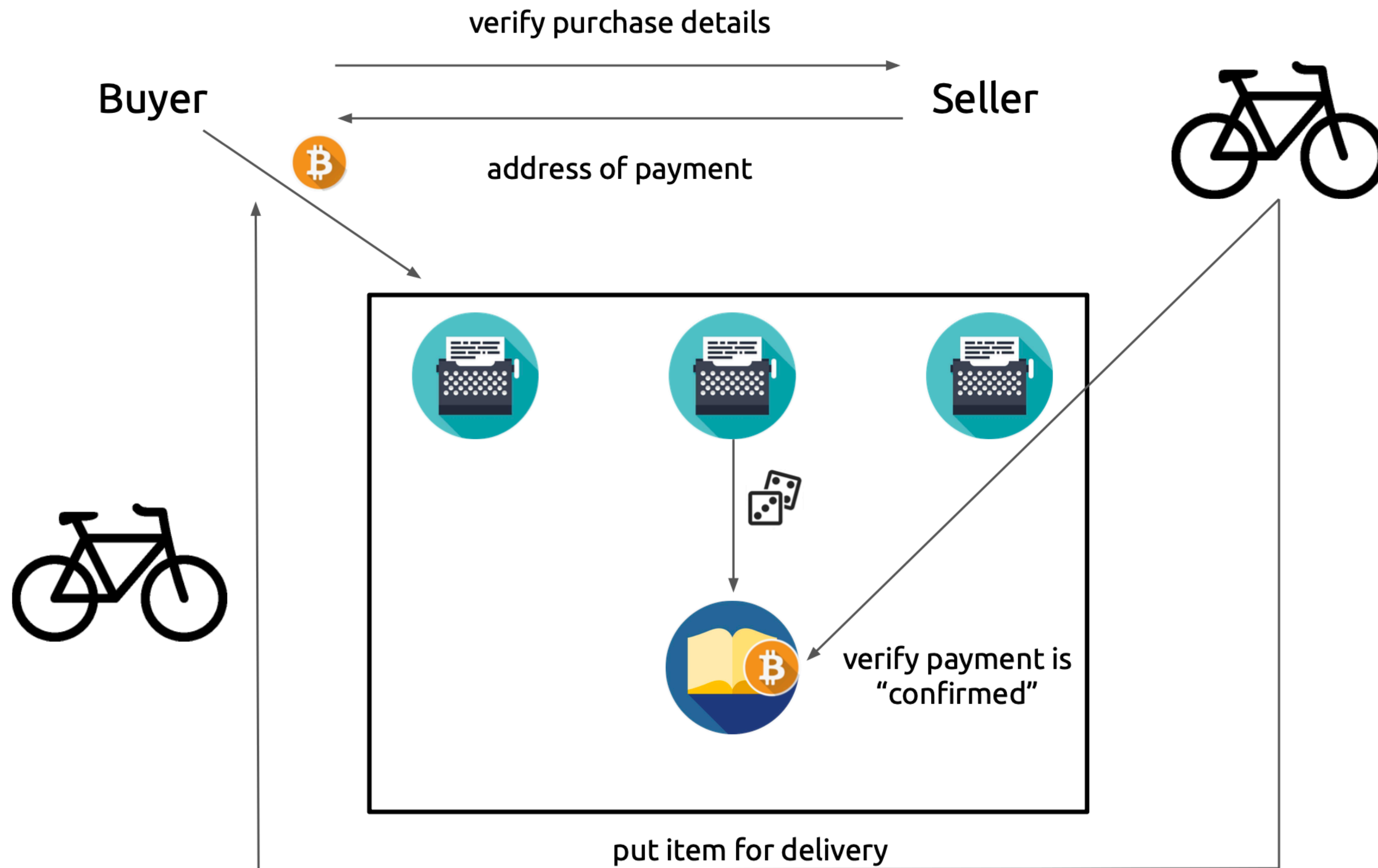
- Each page needs a special combination from a set of dice to be rolled.
- The probabilistic nature of the process is paramount to its security.



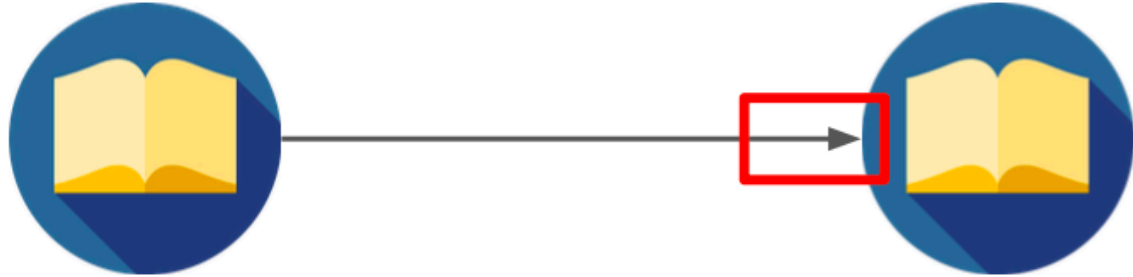
# Never Ending Book: Power

- Anyone can be a scribe for the book.
- As long as one has a set of dice.
- The more dice one has, the higher the likelihood to produce the winning combination to make a page.

# Never Ending Book: Using it



# From Never Ending Book to Blockchain

|                                                                                       |                                                                                   |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|    | The “blockchain”                                                                  |
|   | “Miners” / Computer systems that organize transactions in blocks                  |
|   | Solving a <b>cryptographic puzzle</b> that is <b>moderate hard</b> to solve       |
|  | Using a computer to test for a solution from a large space of candidate solutions |

# Bitcoin

- First decentralized digital currency
- Digital coins you can send through the Internet



## Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto

October 31, 2008

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### Abstract

A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

### 1. Introduction

Commerce on the Internet has come to rely almost exclusively on financial institutions serving as trusted third parties to process electronic payments. While the system works well enough for most transactions, it still suffers from the inherent weaknesses of the trust based model. Completely non-reversible transactions are not really possible, since financial institutions cannot avoid mediating disputes. The cost of mediation increases transaction costs, limiting the minimum practical transaction size and cutting off the possibility for small casual transactions,

# Bitcoin

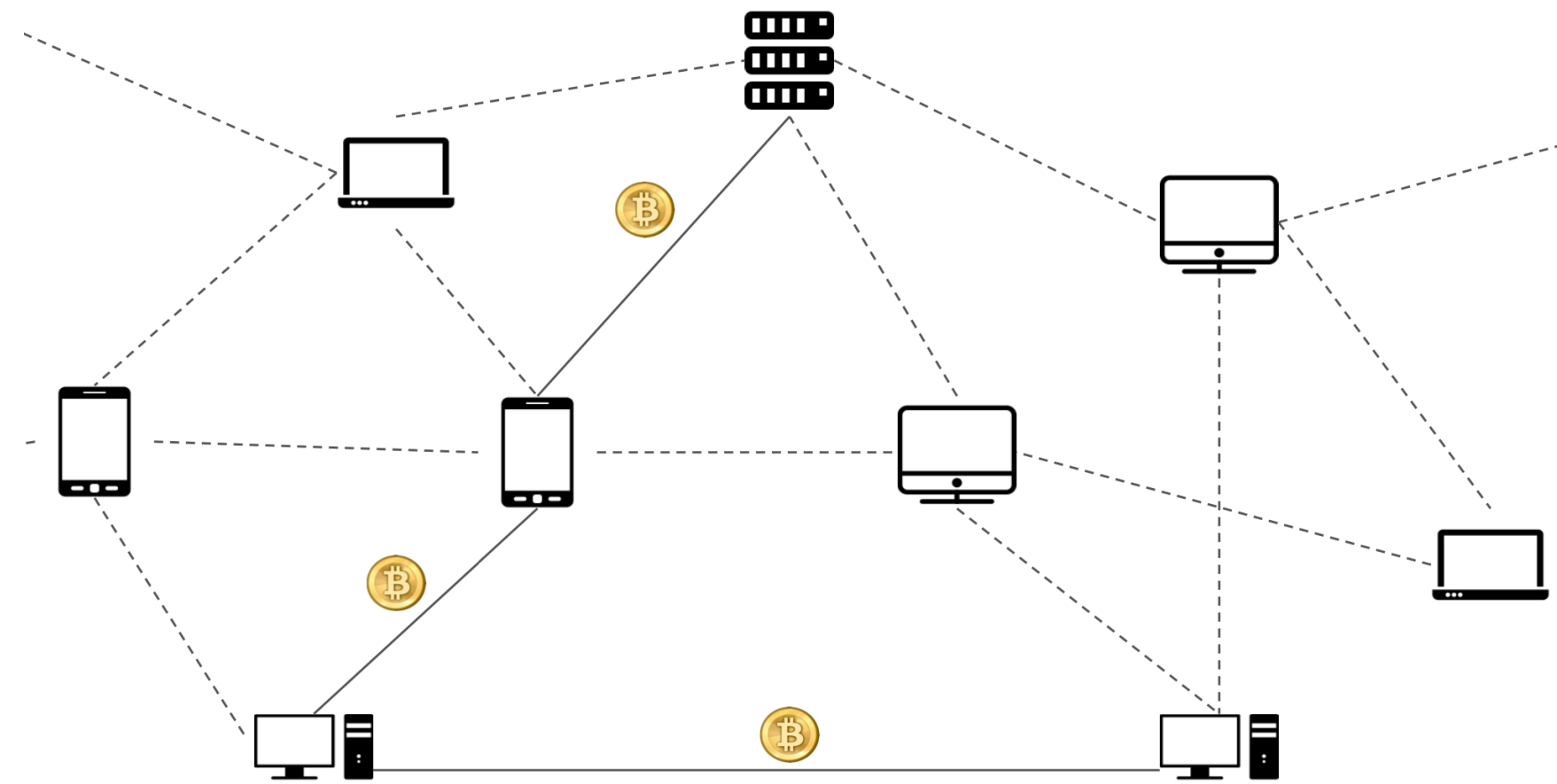
- This is a technical paper published online in the year of the financial crisis.
- **Satoshi Nakamoto** wrote a paper in 2008 about "Bitcoin: A Peer-to-Peer Electronic Cash System".
  - **To-Do: Read the paper!**
- The **real identity** of Satoshi Nakamoto remains **unknown**. We don't know whether it was a single person or a group.
- In **January 2009** the **first block** of the Bitcoin blockchain was **mined**.

# Bitcoin: History



# Bitcoin: Advantages

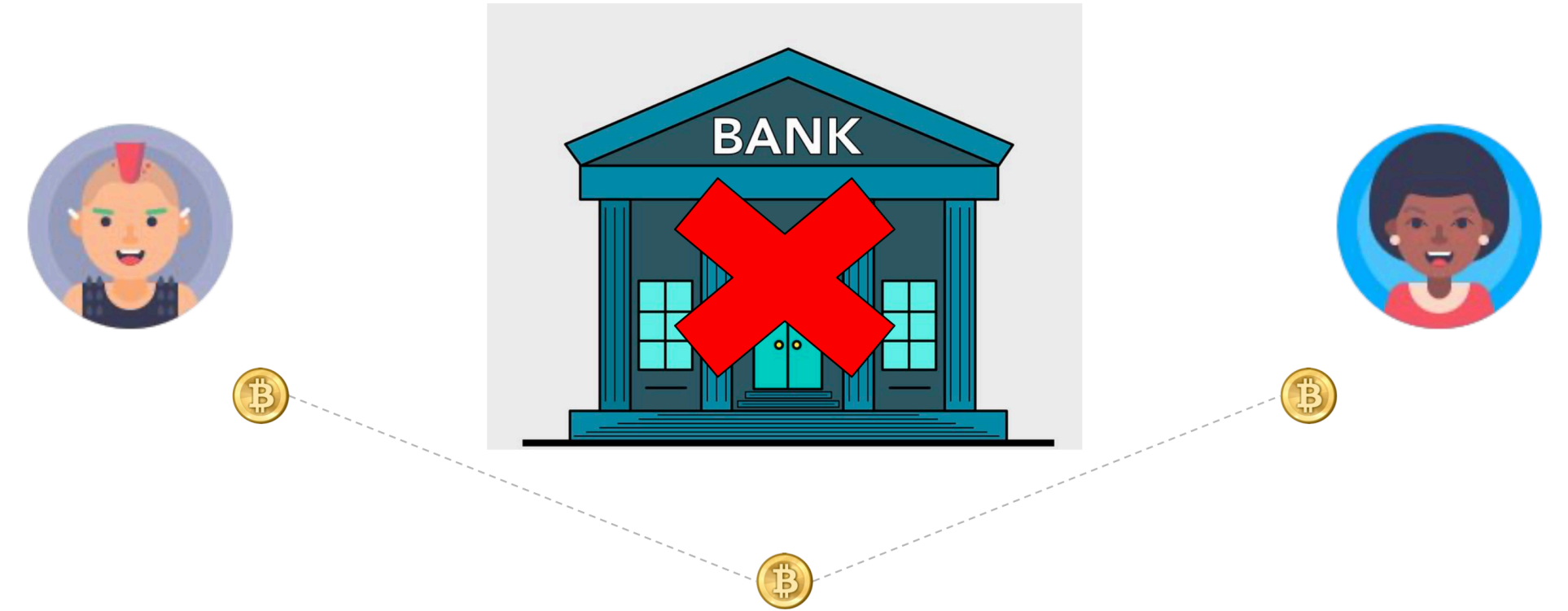
- **Trust-Free:** The system does not require a third party which controls or maintains the system.



- **Tamper-Resistant:** The system is resistant to manipulation. The history of events cannot be changed.
- **Transparent:** Every participant of the system can read and validate all information and the current state.

# Bitcoin: Advantages

- Person-to-Person (Peer-to-Peer, P2P)

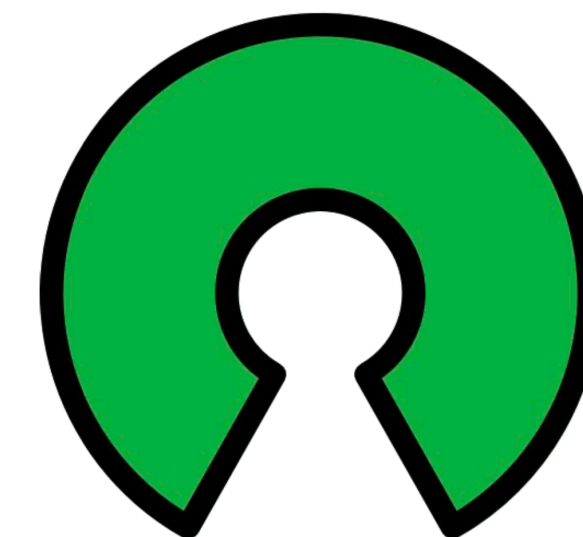
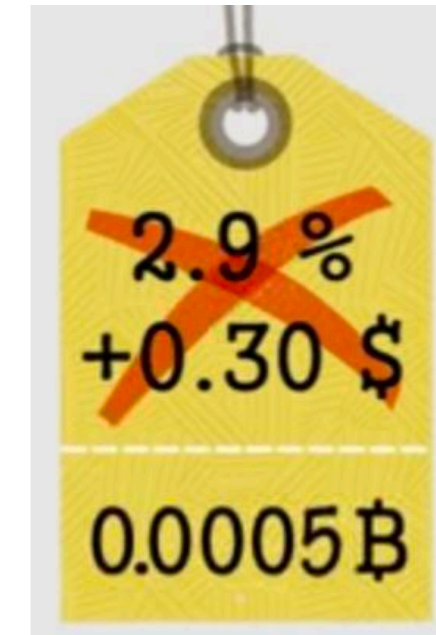


- You own your account



# Bitcoin: Advantages

- Fees determined by free market
- Available to whole world
- Open source: Anyone can review your code

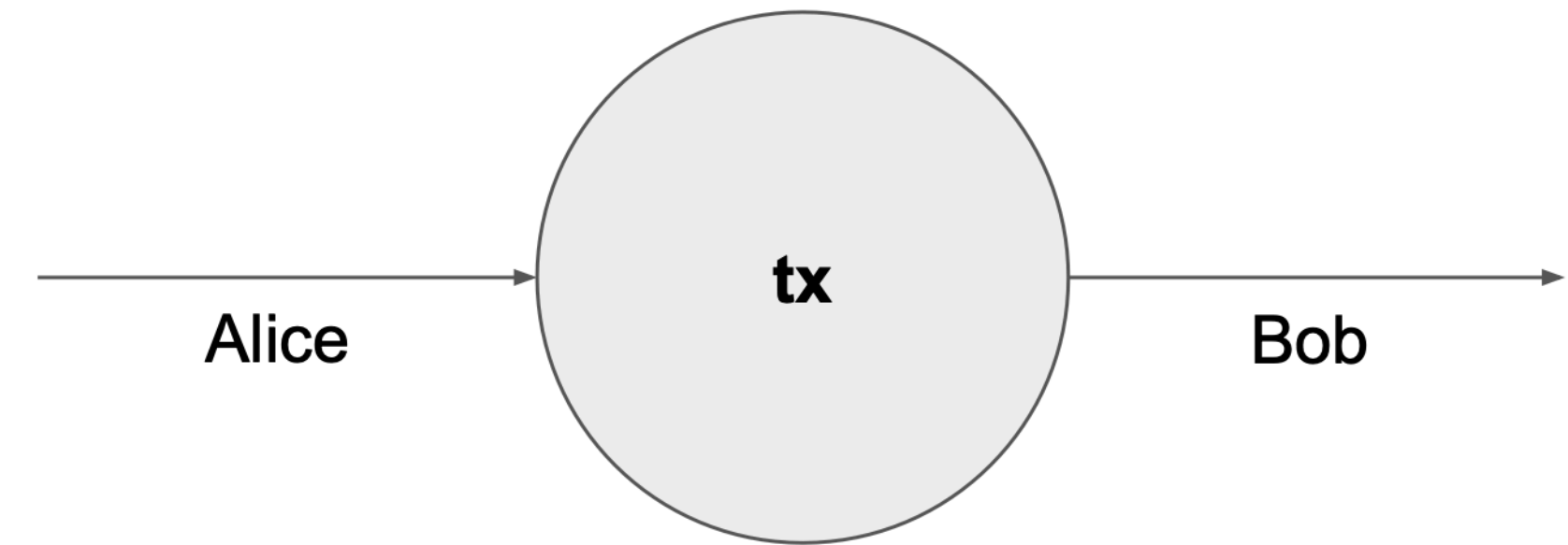


# Bitcoin: Addresses

- Like an email address/IBAN.
- You send bitcoins to a person by sending bitcoins to one of their addresses.
- You can have as many addresses as you want.
- No need to be online to create an address.
- Pseudonymous: A unique address should be used for each transaction.
- Most wallets do it automatically.

# Bitcoin: Transaction

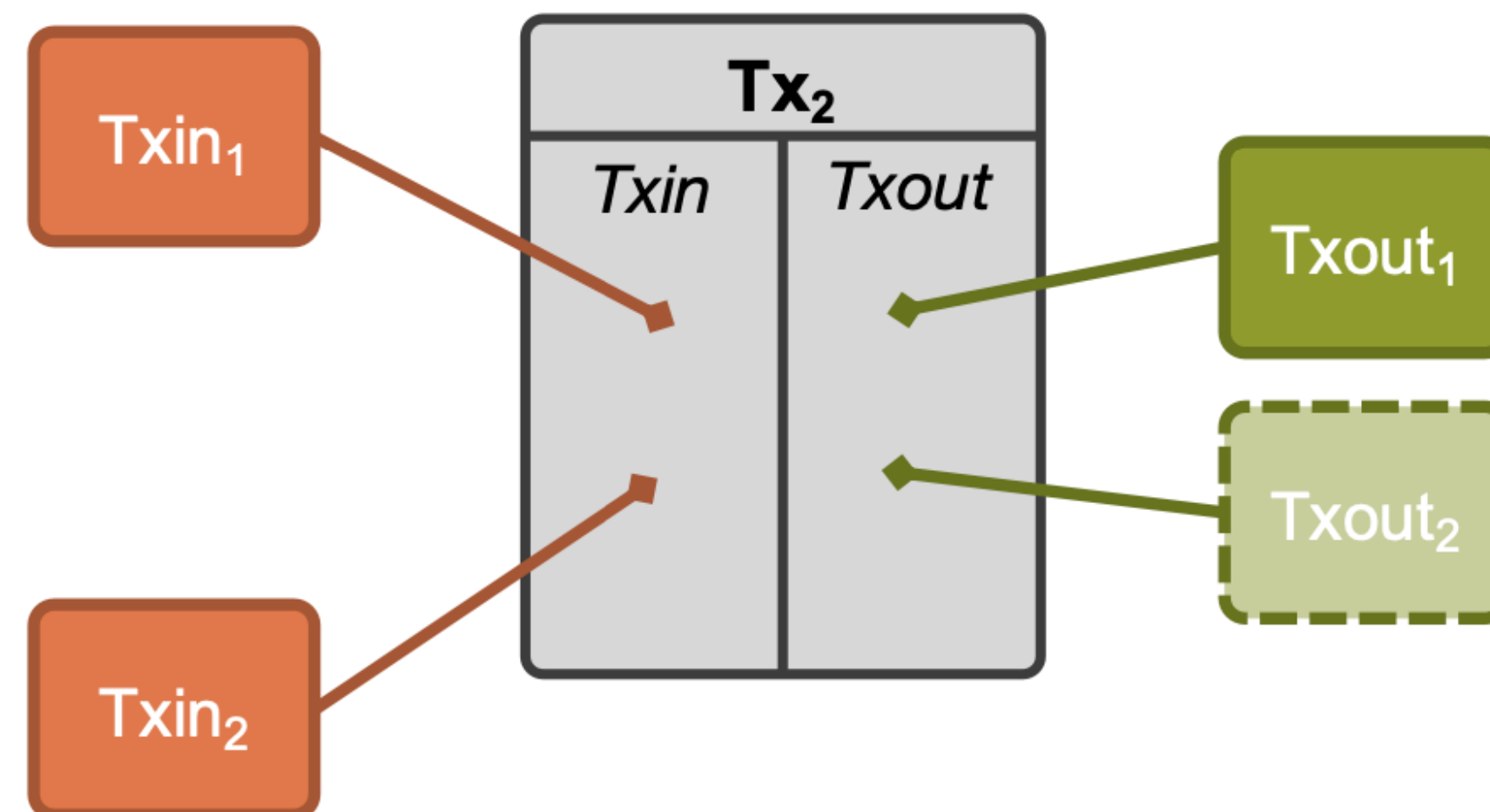
- A simple transaction for financial data
- **UTXO** = “unspent transaction output”



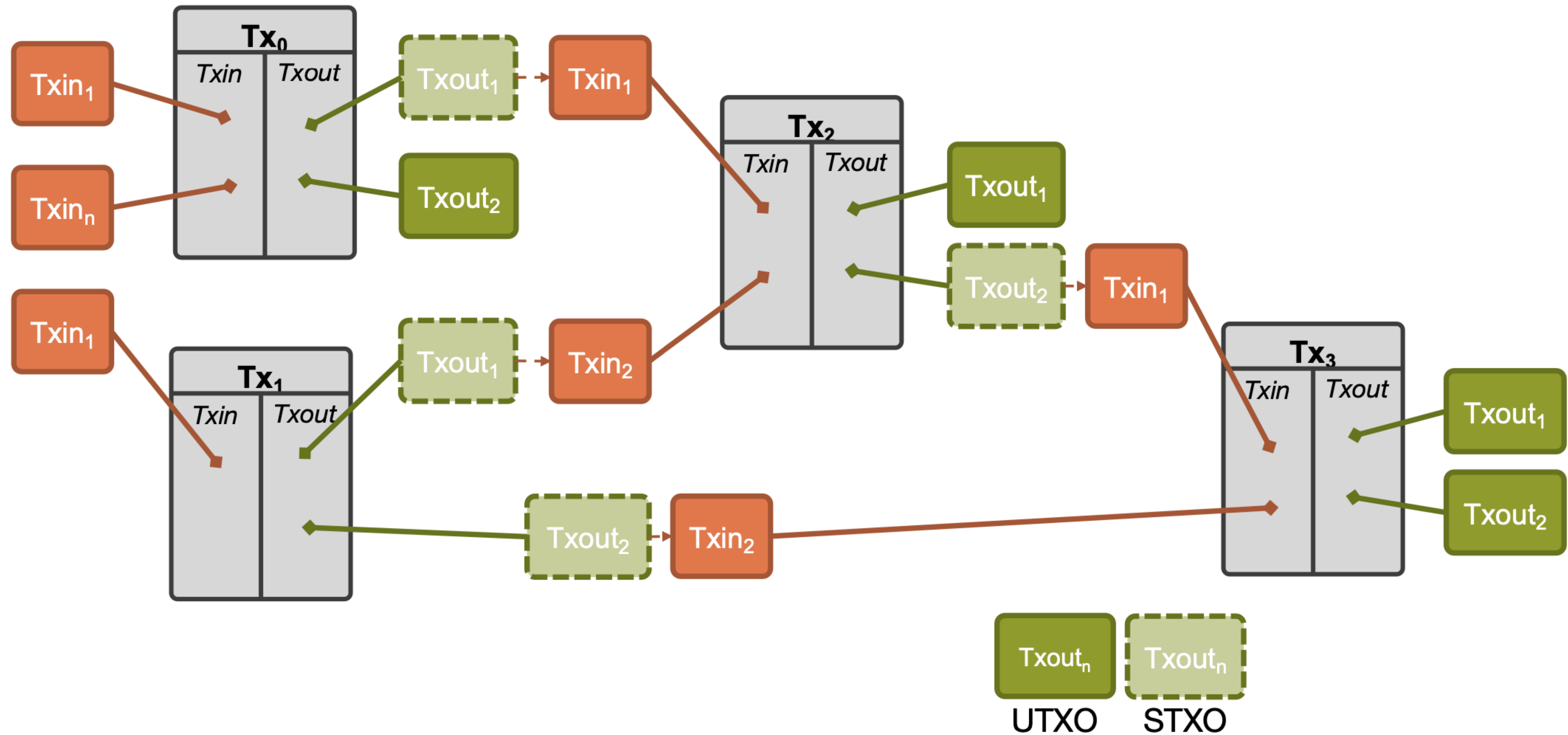
| BLOCKS                                                                    |  | TRANSACTIONS |                |              |
|---------------------------------------------------------------------------|--|--------------|----------------|--------------|
| Transaction Hash                                                          |  | Age          | Amount (BTC)   | Amount (USD) |
| <a href="#">7dd6b6e07ea48577ce11fd43cbf20e259d187defc0888eaa698d7...</a>  |  | 5 seconds    | 1.91072766 BTC | \$7,304.54   |
| <a href="#">94613360083b2e9bdf659d026021c3df9abad4820c1f2bb6add...</a>    |  | 3 seconds    | 0.02130671 BTC | \$81.45      |
| <a href="#">bbda790399d9f44f25d247ea2785b9a687b714665b1fb021cd537...</a>  |  | 3 seconds    | 1.23166111 BTC | \$4,708.53   |
| <a href="#">5d96b437de67fc604b025671f4fa199832b60fc21aedcf94b0455...</a>  |  | 2 seconds    | 0.05533534 BTC | \$211.54     |
| <a href="#">6e7e9284d3c45111a036dab93aae7f7b057e76935c6186051cf92d...</a> |  | 2 seconds    | 0.03158347 BTC | \$120.74     |

# Bitcoin: Transaction

- Transactions (**Tx**) have a number of inputs and a number of outputs.
  - **Inputs (Txin)**: Former outputs, that are being consumed
  - **Outputs (Txout)**: Creation of new coins and transfer of coin ownership
- **UTXO** = "unspent transaction output"
- In transactions where **new coins** are created, **no Txin** is used (no coins are consumed).



# Bitcoin: Transaction



# Bitcoin: Transaction Validation

- Whenever a transaction is validated, it **must pass a long list of checks**:
  - Check syntactic correctness
  - Make sure neither in or out lists are empty
  - Check that size in bytes  $\leq$  MAX\_BLOCK\_SIZE
  - Each output value, as well as the total, must be in legal money range
  - No input maybe a coin base transaction input
  - Check that each input:
    - references an old output
    - references an old output not yet referenced in another transaction
    - contains the correct authorization to spend that old output
    - satisfies the conditions defined by the script
  - Check that the sum of output values  $\leq$  sum of input values
  - Add to transaction pool
  - Relay transaction to peers